

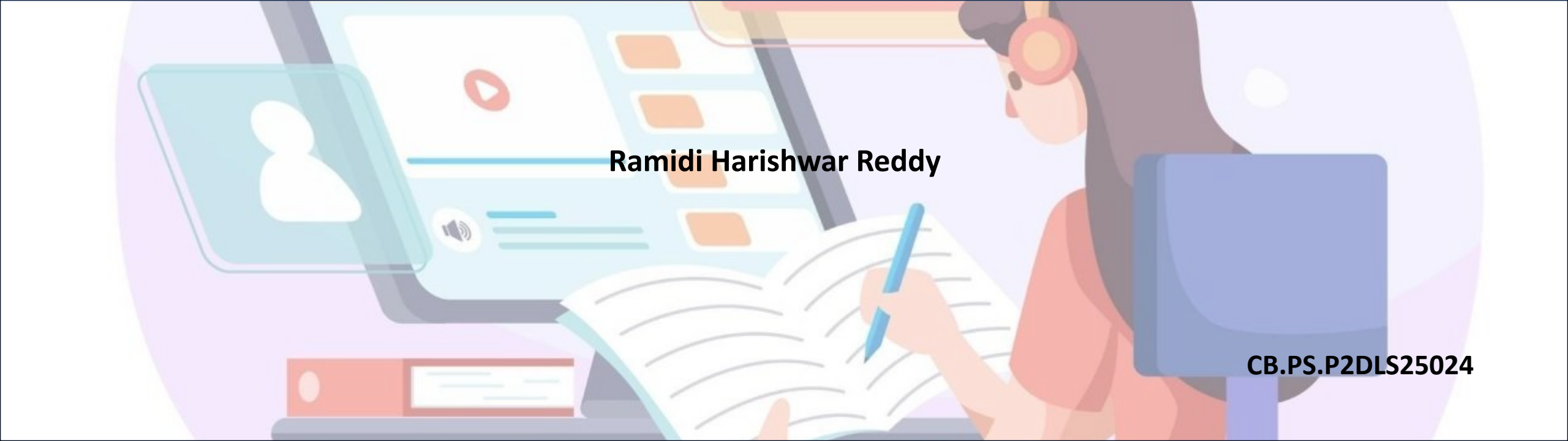
# Online Course

## RECOMMENDATION SYSTEMS

ADVANCE DATA MINING

CASE STUDY

24DLS512



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# INTRODUCTION

This project focuses on developing an Online Course Recommendation System that intelligently suggests relevant courses to users based on their preferences, interests, and learning behavior.

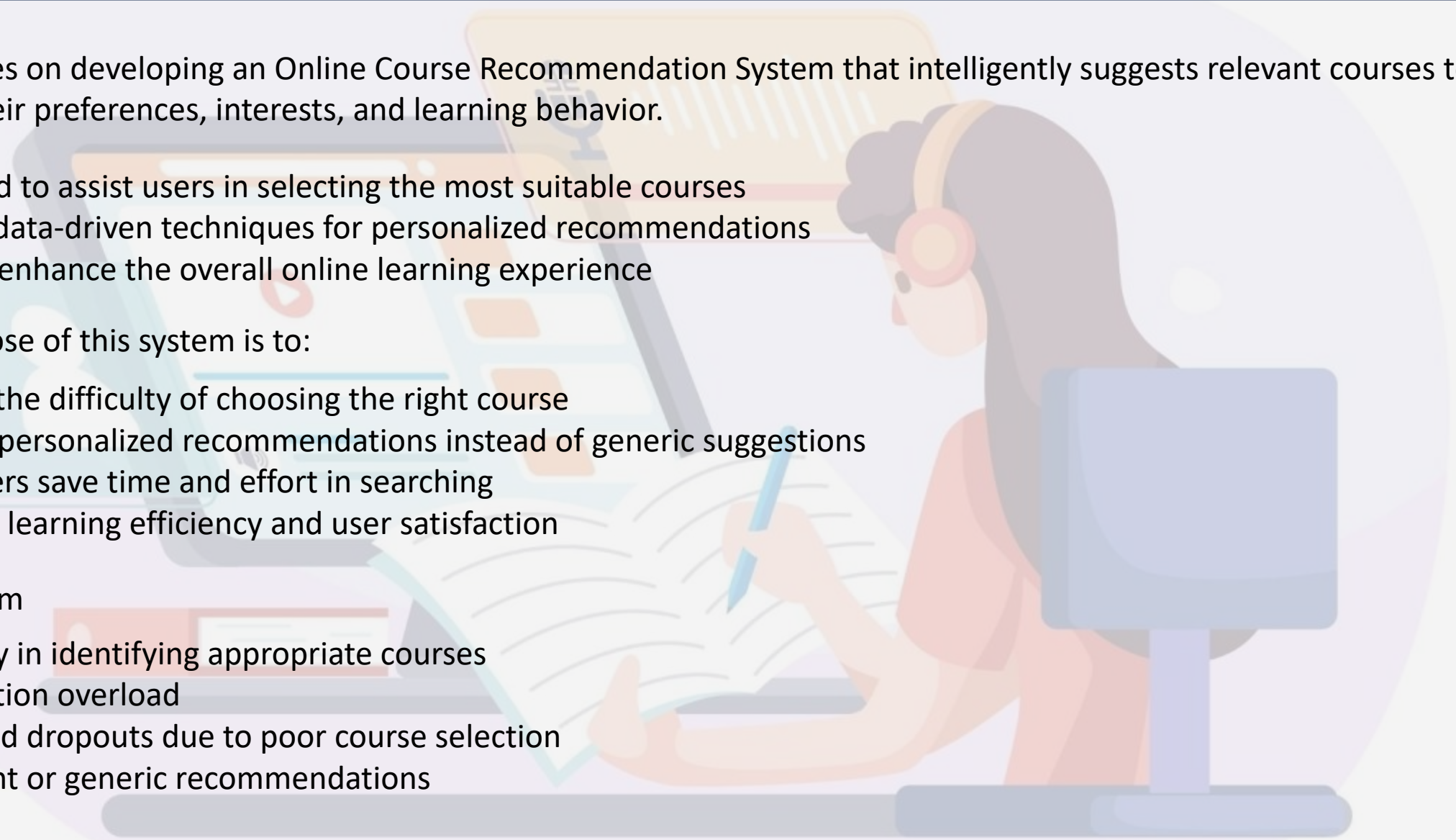
- Designed to assist users in selecting the most suitable courses
- Utilizes data-driven techniques for personalized recommendations
- Aims to enhance the overall online learning experience

The primary purpose of this system is to:

- Reduce the difficulty of choosing the right course
- Provide personalized recommendations instead of generic suggestions
- Help users save time and effort in searching
- Improve learning efficiency and user satisfaction

Real-World Problem

- Difficulty in identifying appropriate courses
- Information overload
- Increased dropouts due to poor course selection
- Irrelevant or generic recommendations



# INTRODUCTION

## How the Project Solves the Problem

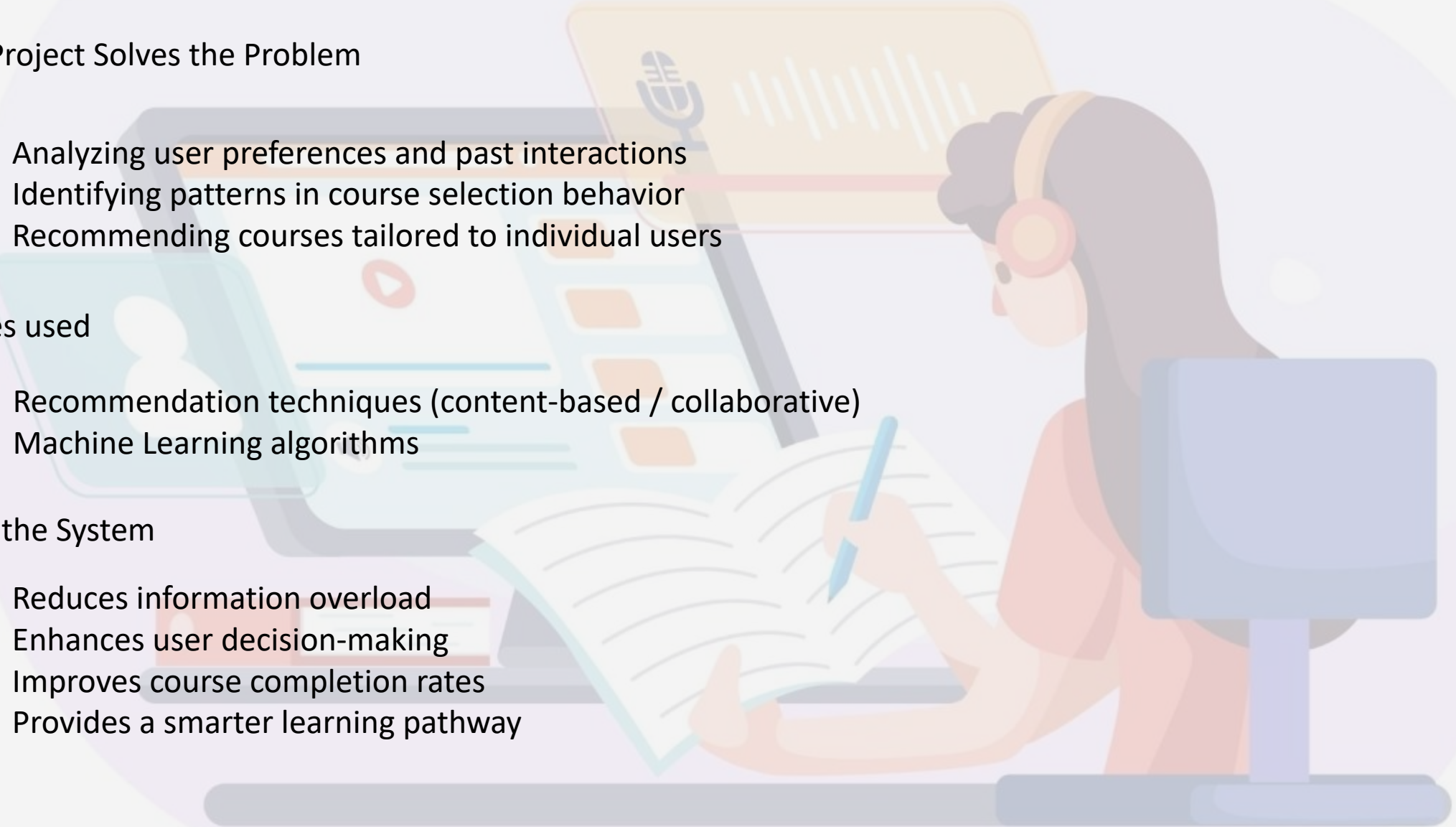
- Analyzing user preferences and past interactions
- Identifying patterns in course selection behavior
- Recommending courses tailored to individual users

## Techniques used

- Recommendation techniques (content-based / collaborative)
- Machine Learning algorithms

## Impact of the System

- Reduces information overload
- Enhances user decision-making
- Improves course completion rates
- Provides a smarter learning pathway



# MOTIVATION, PROBLEM STATEMENT, OBJECTIVES

## Motivation

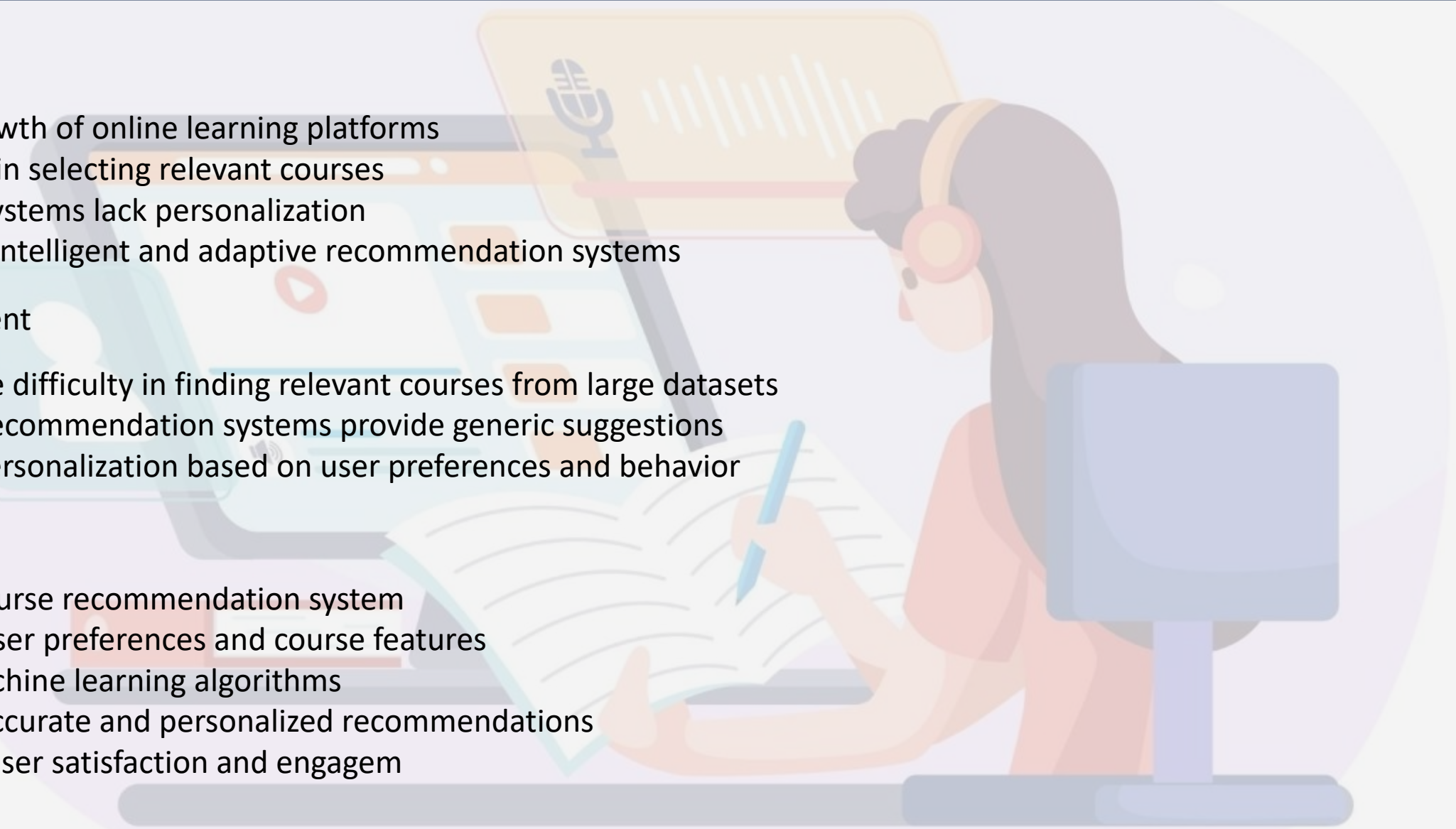
- Rapid growth of online learning platforms
- Difficulty in selecting relevant courses
- Existing systems lack personalization
- Need for intelligent and adaptive recommendation systems

## Problem Statement

- Users face difficulty in finding relevant courses from large datasets
- Existing recommendation systems provide generic suggestions
- Lack of personalization based on user preferences and behavior

## Objectives

- Build a course recommendation system
- Analyze user preferences and course features
- Apply machine learning algorithms
- Provide accurate and personalized recommendations
- Improve user satisfaction and engagement



# DATASET AND PREPROCESSING DESCRIPTION

## Data Extraction & Structuring

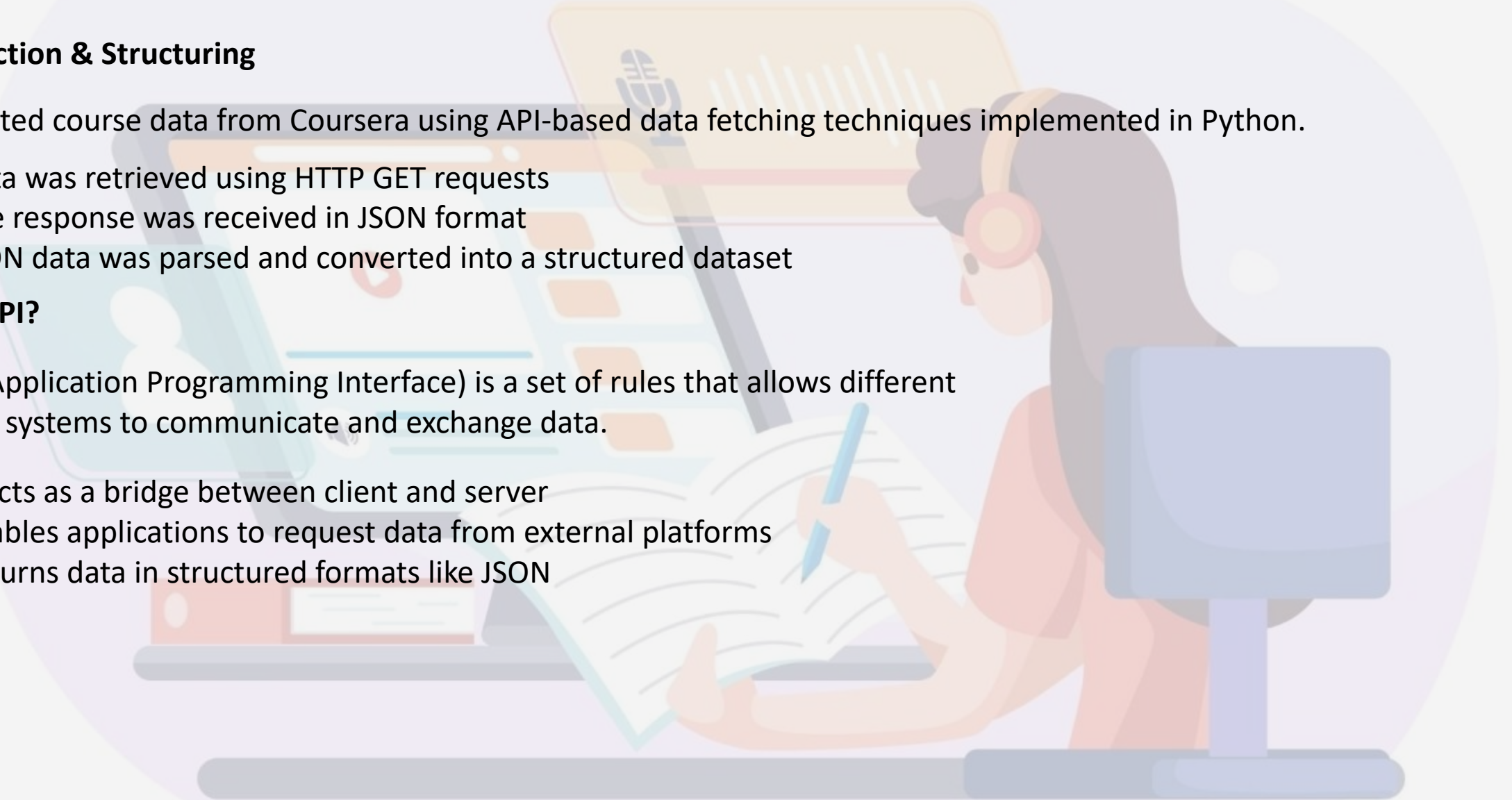
We extracted course data from Coursera using API-based data fetching techniques implemented in Python.

- Data was retrieved using HTTP GET requests
- The response was received in JSON format
- JSON data was parsed and converted into a structured dataset

## What Is API?

An API (Application Programming Interface) is a set of rules that allows different software systems to communicate and exchange data.

- It acts as a bridge between client and server
- Enables applications to request data from external platforms
- Returns data in structured formats like JSON





# DATASET AND PREPROCESSING DESCRIPTION

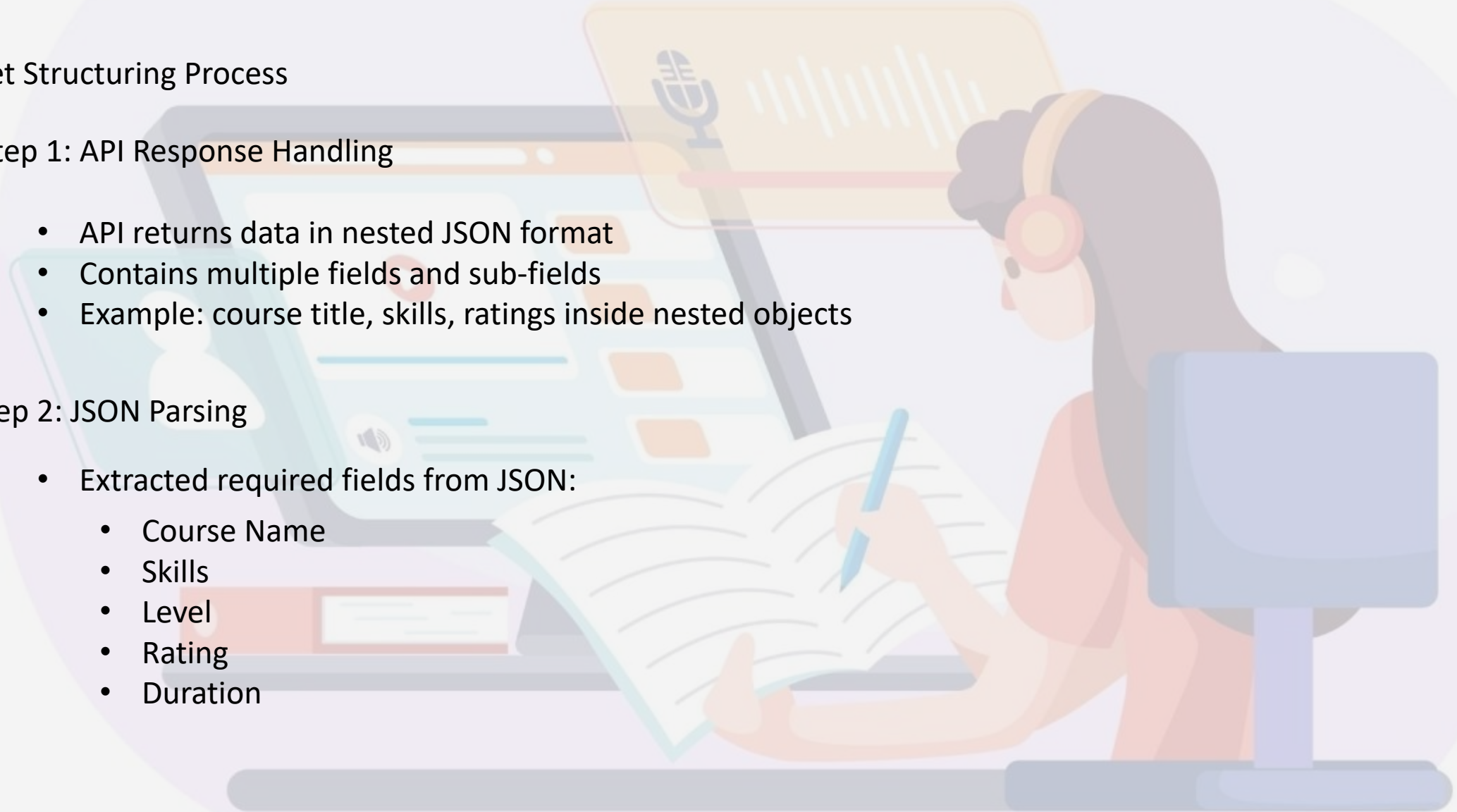
## Dataset Structuring Process

### ➤ Step 1: API Response Handling

- API returns data in nested JSON format
- Contains multiple fields and sub-fields
- Example: course title, skills, ratings inside nested objects

### ➤ Step 2: JSON Parsing

- Extracted required fields from JSON:
  - Course Name
  - Skills
  - Level
  - Rating
  - Duration



# DATASET AND PREPROCESSING DESCRIPTION

## ➤ Step 3: Data Transformation

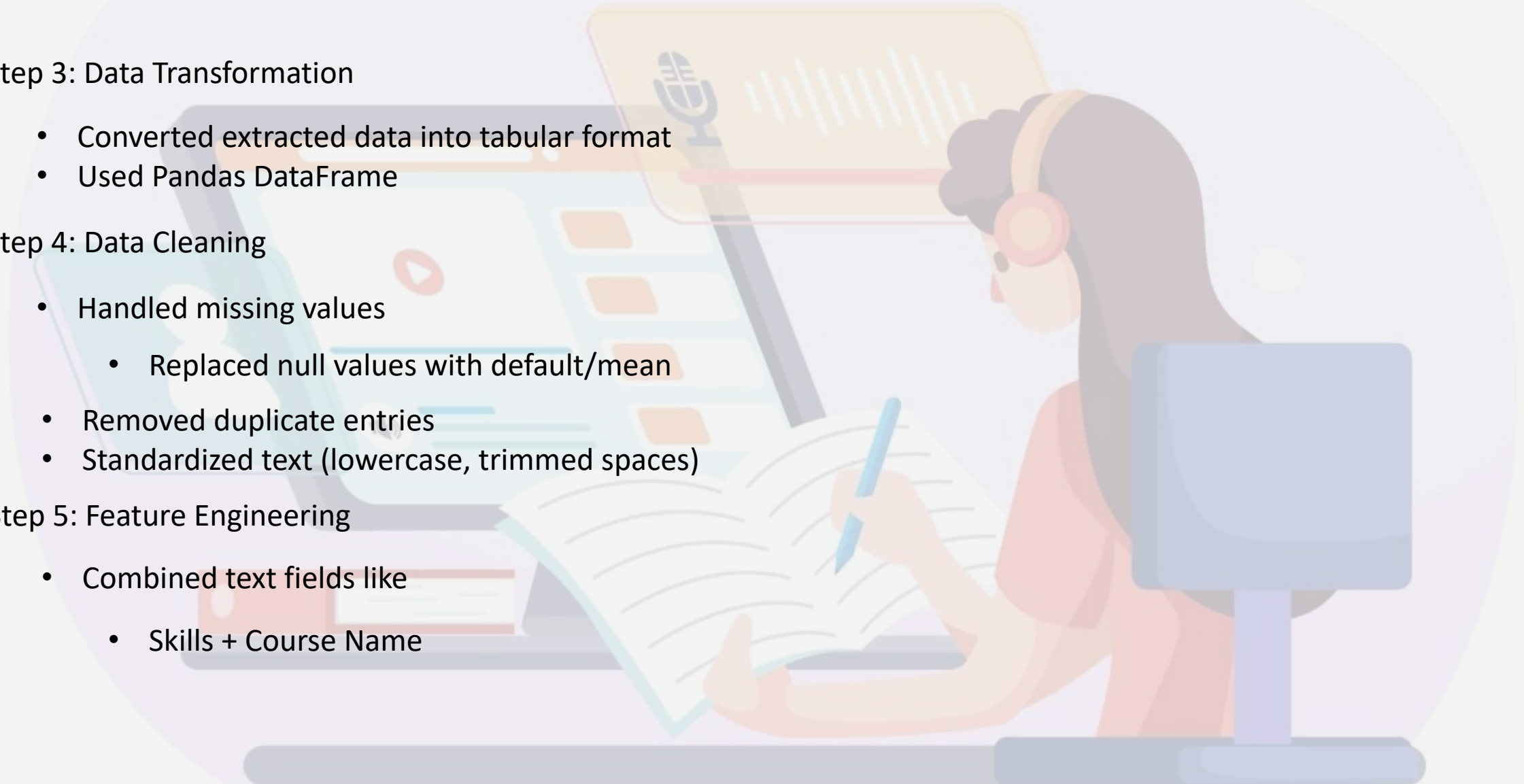
- Converted extracted data into tabular format
- Used Pandas DataFrame

## ➤ Step 4: Data Cleaning

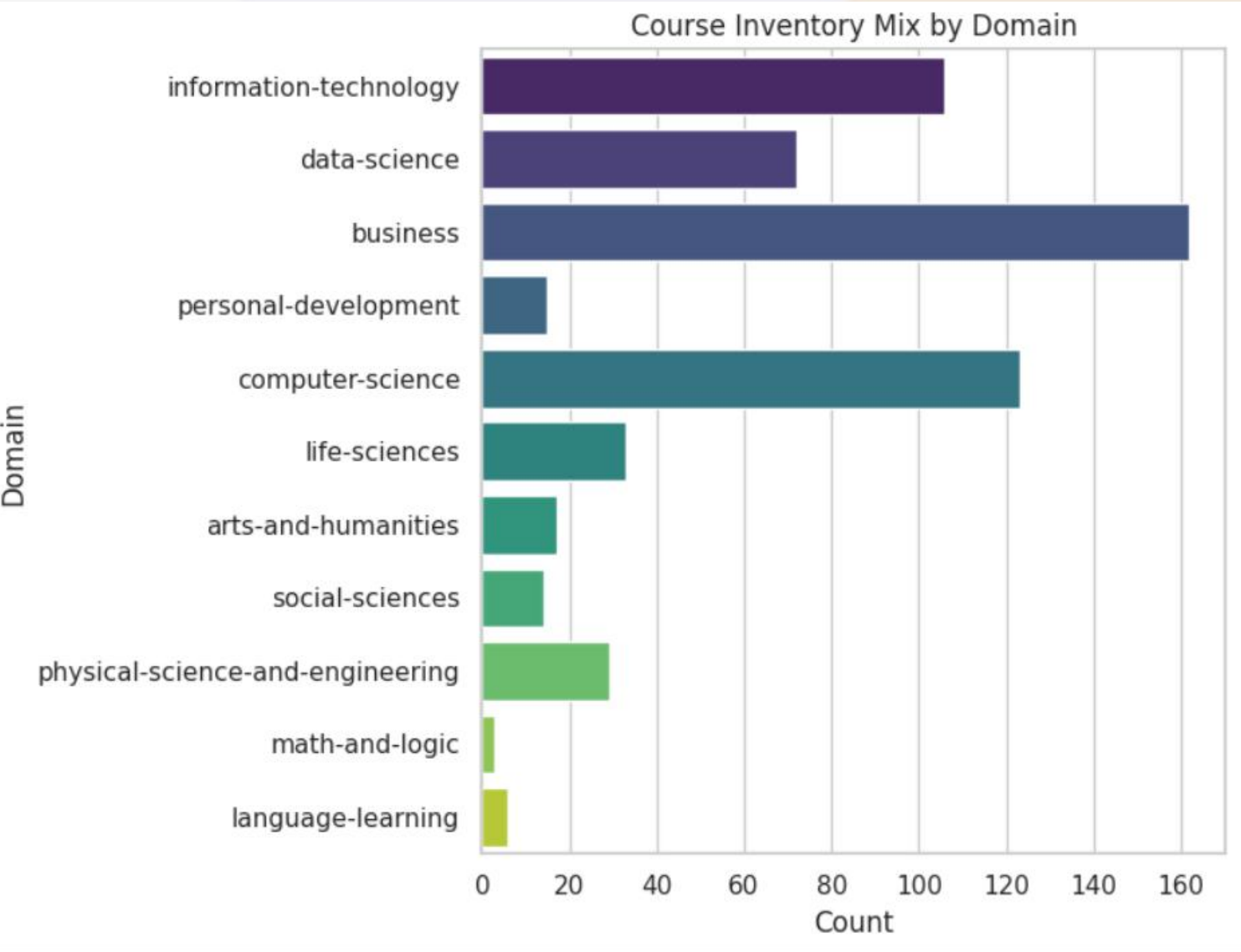
- Handled missing values
  - Replaced null values with default/mean
- Removed duplicate entries
- Standardized text (lowercase, trimmed spaces)

## ➤ Step 5: Feature Engineering

- Combined text fields like
  - Skills + Course Name



# EDA (EXPLORATORY DATA ANALYSIS)



This chart illustrates the distribution of a course inventory across eleven different domains, showing a clear concentration in professional and technical fields. Business stands out as the most dominant category, boasting over 160 courses, followed closely by Computer Science (approx., 123) and Information Technology (approx., 105). Together, three domains make up the vast majority of the offerings. Data Science also maintains a significant presence with roughly 72 courses.

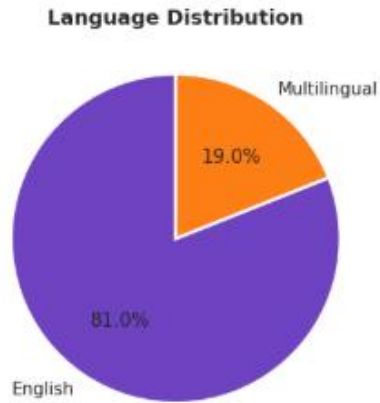


# EDA (EXPLORATORY DATA ANALYSIS)



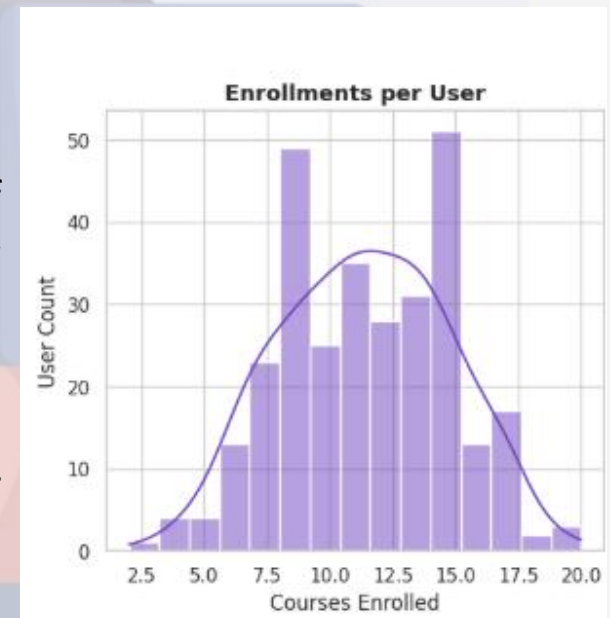
This bar chart illustrates the top 10 most enrolled courses (simulated) based on enrollment activity. Each course in the chart has a similar number of enrollment events, indicating that these courses are among the most popular and highly demanded by users. The courses span a variety of domains, including data structures, artificial intelligence, design, policy, and science, showing that user interest is not limited to a single field but distributed across multiple disciplines.

# EDA (EXPLORATORY DATA ANALYSIS)

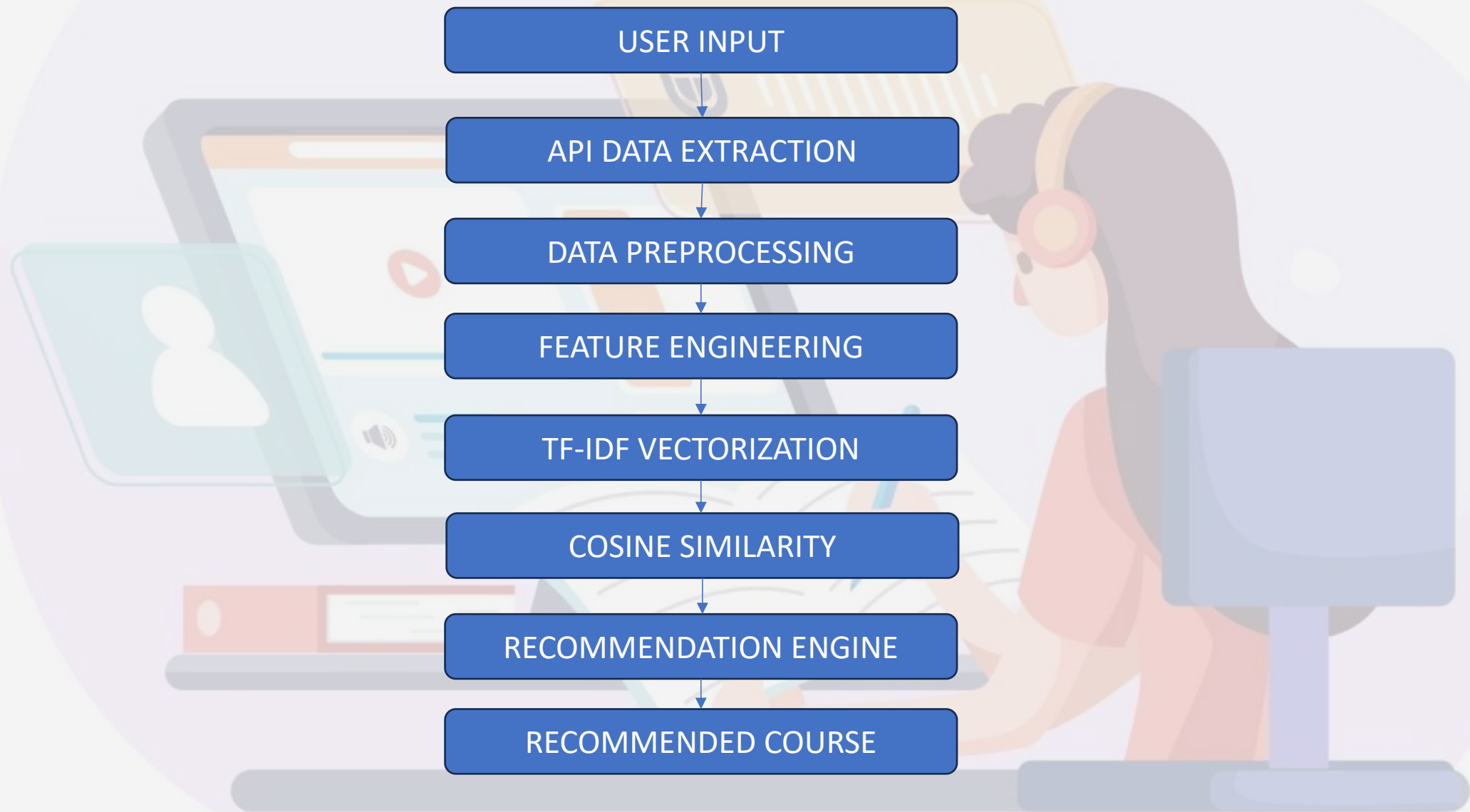


This Pie chart shows the Language Distribution for a set of content or users. English is the overwhelming majority, accounting for 81% of the total. The remaining 19.0% is categorized as Multilingual, indicating a strong primary focus on English-Language materials or speakers with this group.

The Histogram shows the distribution of Enrollments per User, representing how many courses various are signed up for. The data follows a roughly (bell-shaped curve) distribution, with a smooth trend line indicating a peak around 11-12 courses per user.



# BLOCK CHAIN DIAGRAM



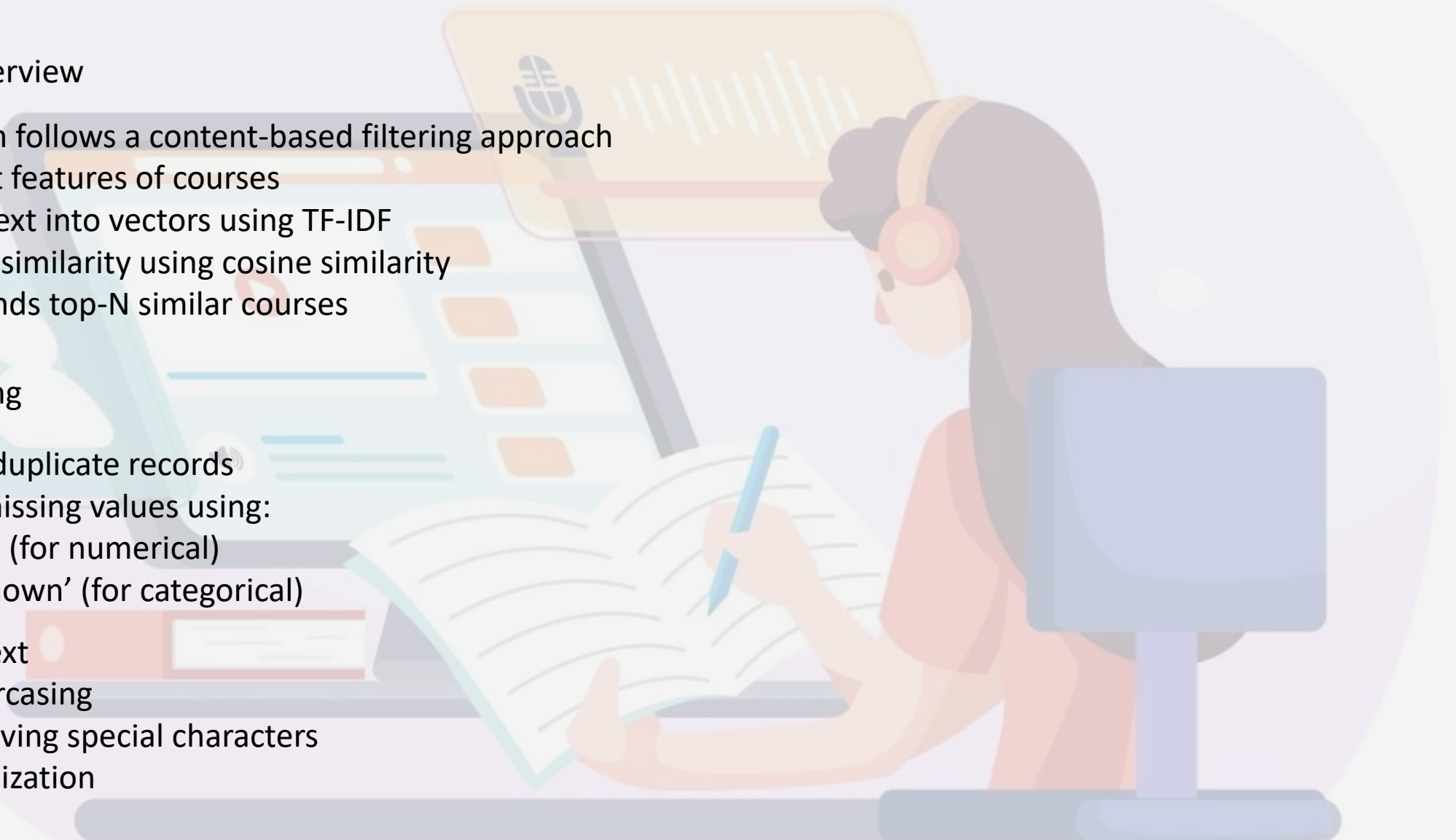
# ALGORITHM /METHODOLOGY

## Methodology Overview

- The system follows a content-based filtering approach
- It uses text features of courses
- Converts text into vectors using TF-IDF
- Computes similarity using cosine similarity
- Recommends top-N similar courses

## Data Preprocessing

- Removed duplicate records
- Handled missing values using:
  - Mean (for numerical)
  - 'Unknown' (for categorical)
- Cleaned Text
  - Lowercasing
  - Removing special characters
  - Tokenization



# ALGORITHM/METHODOLOGY

## Feature Engineering

- Combined multiple features:
  - Course Name (weight: high)
  - Category (weight: medium)
  - Description (weight: low)

## TF-IDF Vectorization

Content:

- Converts text into numerical vectors
- Measures importance of words
- Reduces weight of common words

## Cosine Similarity

- Measures similarity between two vectors
- Value range: **0 to 1**
  - 1 → identical
  - 0 → completely different

## Recommendation Algorithm

- User selects a course
- Convert it into TF-IDF vector
- Compute similarity with all courses
- Sort similarity scores
- Recommend top-N courses



# DEMO

## INPUT

### Smart Course Recommender Portal

Course: Kubernetes Fundamentals



Get Recommendations

## OUTPUT

Currently Viewing

Kubernetes Fundamentals  
INFORMATION TECHNOLOGY

english

✦ Similar Courses You Might Like

Certified Kubernetes Administrator (CKA): Unit 1

INFORMATION TECHNOLOGY

english

Distributed Load Testing Using Kubernetes

COMPUTER SCIENCE

english

Deploy, Manage, and Orchestrate Your Models

INFORMATION TECHNOLOGY

english

Cloud Native, Microservices, Containers, DevOps and Agile

COMPUTER SCIENCE

english

# CONCLUSION

This project successfully developed an Online Course Recommendation System that provides personalized course suggestions using machine learning techniques.

- Real-time course data was extracted using API-based methods
- Data was preprocessed and structured for analysis
- TF-IDF was used for feature extraction from textual data
- Cosine similarity was applied to identify similar courses
- The system generates relevant and accurate recommendations
- The proposed system effectively addresses the problem of information overload by helping users find suitable courses quickly and efficiently.

This system demonstrates how machine learning can be effectively used to solve real-world problems in online education by delivering intelligent and personalized recommendations.”

